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Faculty of Engineering, Environment and Computing School of Computing, Electronics and Mathematics

Analysing of Motor Characteristics for Shell ECO Marathon Project

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Final Thesis-7011CEM: Individual Project

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ABSTRACT:

The proposed project will focus on experimental estimation of the key electrical and magnetic properties of a Permanent Magnet Synchronous Motor (PMSM) to enhance better modelling, prediction of the torque, and control behaviour of the electric drive. It was done in an organised manner, with literature analysis and lab work on experimental aspects to measure stator resistance, Dq and Lq-axis inductance, back-EMF performance, and magnet flux connection. Research was also provided by recent studies carried out on the characterisation of motor and model-based control strategies. They provided the decision on the testing methods and the criteria used to validate this study.

The experimental stage involved test equipment like an LCR meter, an oscilloscope, an inverter supply and manual rotor excitation. The standardised tests included the DC resistance test, the increase in inductance test in UV coil pairing and the no-load back-EMF test, which were compared with the superior methods in the literature. The results obtained a consistent set of parameters of Field-Oriented Control (FOC) modelling. They stressed the impact of the magnetic saturation, cross-coupling and the measurement error on the accuracy of PMSM parameter estimation.

The aspect of critical reflection of test constraints, safety concerns, data validity, time management, and equipment limitations is also part of the project. Ethical, social and legal issues that were assessed during the process were safe electrical operation, responsible data management and academic integrity. Overall, the paper is professionally competent in the application of research-based approaches to extract a validated set of PMSM parameters that can be implemented in simulation, performance prediction and future controller development of electric vehicle and renewable-energy drivers

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LIST OF ABBERIVATIONS:

Abbreviation	Meaning
PMSM	Permanent Magnet synchronous motor
FOC	Field-Oriented Control
Rs	Stator resistance
Ld	d-axis inductance
Lq	q-axis inductance
Ψf	Flux linkage
EMF	Electromotive force
RMF	Rotating magnetic field
EV	Electric vehile
SEM	Shell Eco-Marathon
RMS	Root Mean square
PWM	Pulse with modulation
RPM	Revolutions per minute.

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I would like to extend my heartfelt gratitude to all those who have supported and guided me throughout the course of this project.

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My thanks also extend to the tech team, whose assistance with laboratory access, equipment setup and safety procedures made the experimental phase both smooth and productive. Their practical knowledge and willingness to help at every step greatly contributed to the successful completion of this project.

I want to acknowledge my colleagues and friends for their constant support and insightful discussions, which often sparked new ideas and perspectives. Their encouragement helped me stay focused, positive and committed from start to finish

1. INTRODUCTION:

Permanent Magnet Synchronous Motors (PMSMs) represent the core of many high-efficiency electric propulsion systems used in light electric vehicles, robotics, renewable-energy applications and research platforms. Their growing adoption is driven by their high torque density, low losses and smooth electromagnetic behavior. However, achieving accurate torque control, speed regulations and high-performance operation depend strongly on how precisely the motors internal parts are known. These parameters, like stator resistances, D-axis and Q-axis inductances, permanent magnet flux-linkage, and back EMF characteristics, directly influence controller design, estimation accuracy and overall system efficiency.

In real-World applications, data sheet values of these parameters are usually inaccessible or unreliable in terms of under the conditions pertinent to the system interest. This is particularly so with the case of hub-type PMSMs like the MXUS XF15R rear-wheel motor as in this project, manufacturing variation, temperature and magnetic saturation can lead to large deviations between nominal values. Since the intended drive train of the shell ECO-Marathon electric vehicle would be based on sound modelling to develop future Field-Oriented control method (FOC), it was critical to acquire a valid and experimentally verified parameter of the motor.

The project aims to establishing the significant electrical and magnetic amount of the MXUS XF15R PMSM through the experimental procedures in the laboratory. The test will include the DC Stator resistances, testing incremental inductances with UV terminal linkage, manual rotor excitation testing on D and Q axis and testing on the no-load back EMF waveform with oscilloscopes to obtain the flux linkage. The process is made on a mixture of research finding, laboratory limitations and safety-related factors that is effectively as possible using the equipment readily available, including the LCR meter, bench power supply, controlled three-phase inverter and tachometer.

The project also combines the project-management planning and critical review of the limitations of the methods with the technical experiments. These parameters will assist in the future modelling, the design of controllers and optimization of the drives of the vehicle under study (SEM) and will also help in development of viable observation of PMSM behavior in a laboratory. In general, this paper offers theoretical and practical basis of the high accuracy PMSM control in the real-life engineering setting.

RESEARCH QUESTIONS:

1. How can accurate PMSM parameter identification contribute to improving the design, tuning, and performance of motor control algorithms in electric vehicle applications?
2. How suitable are the experimentally identified parameters for supporting future Field-Oriented Control (FOC) development and modeling for EV applications.
3. How reliably can the key parameters of the MXUS XF15R PMSM (Rs, Lq, Lq and BACK-EMF) be measured using laboratory-based offline testing methods?
4. To what extend do the measured parameters align with theoretical expectations for a surface-mounted PMSM.

OBJECTIVE:

- To measure experimentally the main electrical characteristics of the MXUS XF15R PMSM, to include stator resistance (Rs), d-axis inductance (Ld), and q-axis inductances (Lq) and the features of back-EMF.
- To determine permanent-magnet flux linkage of the motor through measured back-EMF data and using suitable analysis techniques.
- To create a validated parameter set, which can be used in correct PMSM modelling and future Field-Oriented Control (FOC) design, which can guarantee a better torque estimation and dynamic performance.
- To apply laboratory tools and measured and measurement techniques, such as LCR meter testing, oscilloscope waveform analysis and controlled excitation of motor windings, in a safe and systematic manner.
- To critically assess the measurement procedures like limitation, and possible sources of error and give recommendations on how accuracy can be enhanced in future identification tests.

Summary of thesis:
This project investigates the electrical characteristics of PMSM type motor to support Field-Oriented Control development for electric drive applications. The work begins by reviewing existing research paper on PMSM modelling, back-EMF behavior and parameter identification. The methodology then the outlines the experimental procedures of the testing that used to determine the key motor parameters such as Rs, Ld, Lq, back-EMF constant and flux linkage. The results present the measured values and it helps to discusses their relevance to motor control performance. Thesis concludes by reflecting on the accuracy of the experimental readings, limitations of the testing setup and it also mention the recommendations for future work such as load testing, inverter-based control validation and torque-speed characteristics.

2 LITERATURE REVIEW:

The interaction between the rotating magnetic field of the stator and the steady magnetic field produced by permanent magnets in the rotor is the basis for the operation of permanent magnet synchronous motors, or PMSMs. PMSMs are more reliable, more efficient, and have a faster dynamic response than traditional induction machines, which makes them ideal for sophisticated drive systems (Krishnan, 2010).

Since it can decouple torque and flux components through coordinate transformation, which is determined by the rotor flux angle, Field-Oriented Control (FOC) is the most widely used control strategy (Zhu and Howe, 2007).

This method preserves high precision and smooth torque output while reducing the controller's computational load. FOC is perfect for industrial motor control applications and electric vehicle applications because it also allows for higher Pulse Width Modulation (PWM) frequencies and enhanced system responsiveness (Lee et al., 2020).

Recent research thus emphasizes the continuous requirement for hardware validation of PMSM models and experimental parameter identification in order to support dependable control performance and fault diagnosis in actual drive systems (Combes et al., 2017; Bianchi et al., 2017).

A number of experimental methods for identifying PMSM parameters have been proposed in recent studies to increase the precision of dynamic modelling and Field-Oriented Control (FOC). Key parameters, including stator resistance, are frequently determined using conventional techniques like the DC resistance test, locked-rotor impedance measurement, and no-load back-EMF testing (R_s), inductances (L_d, L_q), and permanent magnet flux linkage (ψ_f) (Lee *et al.*, 2020).Flux linkage and incremental inductances under saturation and cross-coupling conditions have been demonstrated to be more accurately estimated using advanced approaches such as high-frequency signal injection and voltage integration techniques (Combes et al., 2017). With these techniques, a small sinusoidal excitation is applied to the stator windings at low speed or at standstill, and the current response is analysed to extract magnetic and inductive characteristics. Researchers created high-precision models appropriate for controller design and real-time PMSM drive monitoring by combining traditional and advanced testing techniques, especially in electric vehicle applications where stability and efficiency are crucial (Zhu et al., 2021).

Broader reviews by Kabalci (2020) and Luna et al.(2017) emphasis the need for the detailed modelling and optimizing the techniques in the microgrid and renewable-hybrid environments. Both studies note the system reliability and control depend heavily on correct parameter values, supporting the methodology implemented in identifying PMSM parameters experimentally. Likewise, Mekhilef, Saidur, and Kaalisarvestani (2018) evaluate the evolution of hybrid systems and highlight the importance of accurate modelling for improving long-term stability of the system.

Chauhan and Saini (2018) examine integrated renewable-energy configurations and identify the modelling challenges associated with varying environmental and load conditions. Its emphasis on robust characterisation is relevant to the challenges faced when identifying PMSM parameters experimentally. Similarly, Gupta and Singh (2022) assess advanced energy-management strategies for hybrid microgrids and show that system stability is tied to high-precision control algorithms. This insight parallels the requirement for accurate Ld, Lq and Ke and Rs values when implementing Field-oriented control methods in PMSMs.

Gore et al.(2021) investigated the use of Field-Oriented Control (FOC) for regulating the speed of a PMSM driven electric vehicle and demonstrated that accurate motor parameters are critical for achieving stable and efficient operation. Their MATLB-based EV model included regenerative braking, torque control and parameter-sensitivity analysis, showing that variation in stator resistance, inductances and flux linkage lead to increase current consumption, reduce battery state-of-charge and a decline in overall drive performance. This work emphasises that PMSM rely heavily on precise d-q axis modelling and accurate parameter identification for effective FOC implementation, reinforcing the importances of experimentally determining Rs, Ld, Lq and ψf as carried out in the present study.

Lara, Xu and Chandra (2016) analysed how rotor position errors affects the performances of Field-Oriented Controlled PMSM drives used in electric vehicle applications, shoeing that even small angular errors significantly increase torque ripples across different operating region of the motor. Their extended dq-axis analytical model demonstrated that inaccurate rotor angle feedback distorts the park transformation, leading to incorrect id-iq regulation and demonstrated torque output-particularly above base speed during flux-weaking operation. Simulation and experimental results on both 400 W and 80 KW PMSM confirmed that torque ripple can rise sharply with position sensor inaccuracy, highlighting the need for precise parameters estimation and error compensation algorithms. This study reinforces the importances of correct flux linkage, inductances and resistance value when implementing robust FOC algorithms, directly supporting the relevance of accurate experimental identification of Rs, Ld, Lq and back-EMF in this project.

Rebelo and Silvestre (2018) presented the design and experimental evaluation of a coreless permanent magnet synchronous motor (CPMSM) developed specifically a Shell Eco-Marathon prototype vehicle. Their working how removing the iron core eliminates hysteresis and eddy-current losses, allowing exceptional high efficiencies above 95%, although at the cost of very low inductances increased susceptibility of current ripple. The authors demonstrated that careful optimisation of stator geometry, Litz wire winding and Halbach magnet arrangement significantly improves torque production and reduces copper losses. Their experimental results also showed that the shape of the Back-EMF waveform strongly depends on the magnet and coil alignment, with deviations from sinusoidal profile affecting controller performance. This study reinforces the importances of accurately characterising PMSM electrical parameters such as Resistance, inductances and flux linkage-since these directly influences the efficiency, torque quality and FOC performance, aligning closely with the objectives of this project.

However, there is a need for this research because few studies have experimentally validated PMSM parameter identification for small-scale electric drive applications like the Shell Eco Marathon prototype.

3 BACKGROUND AND THEORETICAL FRAMEWORK:

3.1 Motor specifications (MXUX XF15R PMSM):

The MXUS XF15R is a generated rear-hub permanent magnet synchronous motor designed for light electric vehicle applications, particularly e-bikes and prototype EV system. It operates at a nominal Voltage of 36V and delivers approximately 350W of rated power, which makes it well-suited for low-speed, high efficiency propulsion. Internally, the motor uses a planetary gear reduction system with a 5:1 ratio, allowing the internal rotor to rotate five times faster than the wheel. This gearing multiplies the torque output to around 1.98Nm at the wheel. This gearing multiplies the torque output to around 9.8Nm at the wheel, significantly improving acceleration and climbing capability despite the compact size of the motor.

The motor contains 20 permanent-magnet poles (10 pole pairs), which determine the relationship between meachincal and electrical speeds and defines the motor's back-EMF frequency. Its estimates no-load speed of 285RPM at 36V algins with specification for typical hub-drive EV application. With an efficiency rating above the 78%, the XF15R offers responsible low copper loss and mechanical losses, making it suitable for energy -efficient electric drive system such as the Shell Eco-Marathon prototype.

Additional features include an IP54 rating for environmental protection, compatibility with V-brake and disk brake system, and hall sensors interface providing one electrical pulse per cycle for rotor position and speed sensing. These characteristics make the XF15R a robust and adaptable PMSM, and its well-defined mechanical and electrical structure forms the foundations for the parameter's identification experiments carried out in the project.

Parameter	Value
Position	Rear
Construction	Geared (internal reduction)
Operating voltage	36V (Used Voltage)
Rated power	350W
Rated speed	~285 RPM @ 36V (Estimates)
Magnet poles	20 poles (10 pole pairs)
Gear ratio	5:1 (output torque = 5*shaft torque)
Shaft torque (Estimated)	`1.96 Nm (derived from output torque/5)
Output torque (Estimated)	~9.8Nm (from P/ω at 36V)
Wheel diameter compatibility	20–29-inch wheels
Brake type	V-brake / disk brake
Efficiency	≥ 78%
IP Rating	IP54

Noise grade	≤ 55dB
Sensor Interface	Hall sensor + senseless compatibility
Speed detection Signal	1 pulse per electric cycle
Cable length/type	250 mm, G9.1 connector
certification	ROHS / CE

3.2 Working principle Of PMSM MUXS XF15R:

The MXUS XF15R is a three-phase permanent magnet Synchronous motor (PMSM) that operates interaction between the rotating magnetic field produced by the stator resistances and the constant magnetic field of the rotor magnets.

When the three-phase AC current flows the stator windings, it creates a rotating magnetic field (RMF). The motor, which has surface-mounted permanent magnets, locks onto this rotating field and begins to rotate at the same speed.

As the stator field rotates, it exerts a magnetic pull on the rotor poles, generating electromagnetic torque that causes the rotor to run continuously. Because the rotor and stator fields remain synchronized, the motor runs

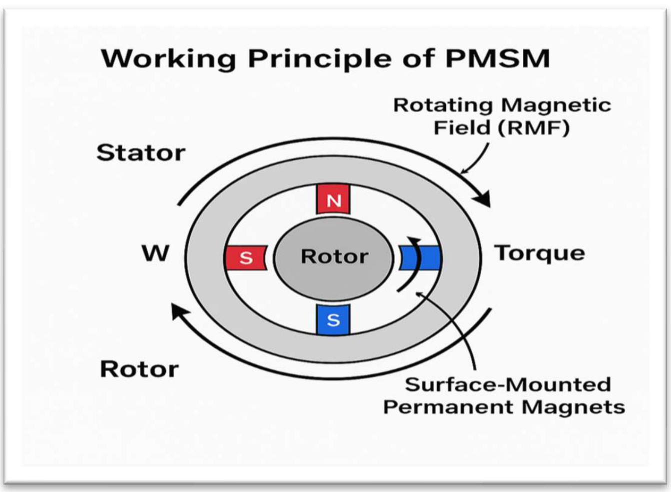


Figure 1 working principle of PMSM

smoothly with no slip and delivers a consistent torque output.

In this motor, the magnetic flux linkage comes from the rotor magnets and the torque(T) by

$$T = {}^{3}/_{2}\, p\psi Iq$$

Where p is the number of poles pairs and lq is the torque producing current.

The MXUS X15R surface-mounted design gives nearly equal inductances along the both d-axis and q-axis (Ld=Lq), resulting in a sinusoidal Back-EMF and low torque ripple.

This smooth performance, quick dynamics responses, and high efficiency make the MXUS XF15R motor ideal for light electric vehicles and energy efficiency applications.

3.3 Importances of PMSM Parameters:

A precise understanding of important electrical and magnetic parameters would be important in modeling and controlling permanent magnet Synchronous Motors (PMSMs). Theses parameters have direct effects on production of the torque, dynamic response, efficiency and stability of advance control strategies like Field-oriented control (FOC). The most essential parameters of the PMSM to be analyzed and controlled are the stator resistances (Rs), D-axis inductance (Ld), Q-axis inductance (Lq) and the flux linkage of permanent magnets (ψf).

Stator resistance (Rs) defines the copper losses in the windings and is directly related to heating, efficiency and current regulations accuracy. Temperature changes in Rs can also introduce considerable error in current control and torque estimation and good measurement is essential to assure stable drive operation.

Ld is the D-axis inductance of the magnetically powered motor, the capacity the motor to store magnetic energy in the direct axis, which is perpendicular to the rotor magnetic field. Ld is important in the weaking of flux and it affects the behavior of the motor as it moves in high speed where voltage constraint is important.

Q-axis inductance Lq characteristics the magnetic response perpendicular to the moving axis, which is the cause of generating electromagnetic torque. Lq has a direct relationship with the current and torque, so fine identification is required to obtain stable and predictable torque when smooth acceleration and high dynamic performance are required.

Permanent- magnet flux linkage, represents the strength of rotor magnets and it determines the Back-EMF as well as the torque constant of a motor. The most important parameters of the PMSM modeling since it characterizes how well an electrical energy could be converted to the mechanical power.

These parameters are basis of correct mathematical models. Accurate determinations of Rs, Ld, Lq and ψf is thus fundamental to the process of simulation modelling, Performance Prediction, as well as the creation of a high-efficiency control algorithm in electric vehicle use.

3.3 Field-Oriented Control (FOC) Overview:

Field oriented control (FOC) is a popular control approach in PMSM control since it allows precise and constant torque generation. FOC operates by converting the three phase stator currents to a rotating Dq-references frame that is rotor flux aligned. This enables the motor current to be differentiated into D-axis current which operates flux and the q-axis current which produce torque directly.

Through decoupling of these parts, FOC has the capability of running motors with smooth and efficient operation and rapid dynamic response of a well-controlled DC motor. The procedure also facilities the fluxweakening, permitting the PMSM to work at greater velocities near voltage constraints, necessary in electric-vehicle use.

Proper parameter of PMSM i.e Rs, Ld, Lq and (ψf) are essential in effective FOC implementation since they affect transformations, regulations of current and estimation of torque. Mistakes in these parameters may decrease efficiency or lead to unsteady behaviour. It is due to the experimental parameter identifications is significant in obtaining credible FOC performance.

PMSM and BLDC type-motors have become the dominant technology in modern electric drives systems due to their high efficiency, fast dynamic response, and wide construability. These motors are currently use across major industries, including electric vehicles, e-bikes and autonomous mobility platforms such as those produced by Bosch, Shimano and segway. In the automotive sector, PMSM drives are used in commercial EVs from Tesla, Nissan, BMW, and Hyundai because of the sinusoidal back-EMF enables smooth torque delivery and high-power density. Similar motor technologies are employed in aerospace applications such as drones, UAV propulsion system and eVTOL aircraft due to their lightweight construction and compatibility with Field-Oriented Control. Industrial automation companies like Siemens, ABB and Mitsubishi rely on PMSM/BLDC motors for robotics, CNC machinery and precision servo systems.

These widespread real-time applications highlight the importance of accurate parameter identification-such as stator resistance, inductance and flux linkage which directly influence efficiency, torque control and overall system performance. The experimental methods used in this project reflect the same engineering practices applied in EV, aerospace and automation industries.

3.4 summary of key concepts:

The following section has described the main principles of the work of MXUS XF15R PMSM, as well as the mechanism of the production of the torque and the importances of such key parameters as Rs, Ld, Lq and ψf to control and model the engine correctly. Field oriented-control (FOC) was briefly described to emphasise on reliance to accurate parameters values. It is this principle that gives the experimentation measurement methods in the methodology section below the required background.

4 METHODOLOGIES:

This chapter outlines the experimental procedures that was used to determine the basic electrical and magnetic properties of MXUS XF15R permanent magnet Synchronous Motor (PMSM). The methodology was structured in such way that it was practical given the current laboratory facilities and produced a set of parameters which could be further modelled using PMSM and develop Field-Oriented control (FOC) strategies.

All tests were performed under strictly regulated laboratory conditions on the MXUS XF15R rear-wheel hub motor and considerations of the safety were also taken into account, as well as the reproducibility.

4.1 Overall Experimental setup:

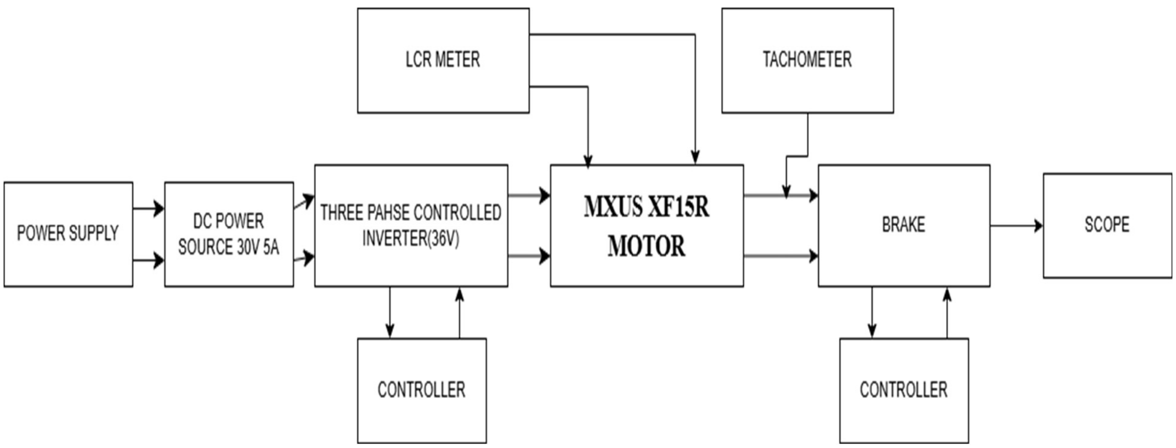


Figure 2Overall experimental setup for MXUS XF15R PMSM testing

1. power supply:

The power supply provides the initial electrical input required to energize the system. It feeds a stable DC voltage to the next stage, ensuring that the inverter receives a clean and reliable input. Without this block the entire setup would not have a controlled source of electrical energy.

2. Dc power source (30v, 5A):

This regulated Dc power source provides the exact voltage and current needed to operate the three-phase inverter safely. It happens prevent voltage fluctuations and protects the motor and electronics from unexpected spikes. Its role is simply to ensure a stable DC link for the inverter.

3. Three-phase controlled inverter:

The inverter is the heart of the motor-driven system. It converts the DC supply into a three-phase waveform that can drive the MXUS XF15R PMSM. By controlling the switching signals, it adjusts motor speed, direction and torque. This block is essential for dynamic testing, FOC implementation and any form of controlled motor operation.

4. Controller (for inverter operation):

The controller generates the PWM signals required by the inverter. It determines how much current flows into the motor and regulates the three-phase output. When implementing Field-Oriented control method (FOC), this controller would calculate the id and iq currents and feed them to the inverter to achieve precise torque control.

5. MXUS XF15R PMSM (Motor under test):

This is the central component of the experimental setup. All measurements are-such as resistance, inductances, back-EMF and dynamic response-are taken from this motor. It interacts directly with the inverter, brake, tachometer and measurement instruments. Understanding the motor's behavior is the main objective of the project.

6. LCR Meter:

The LCR meter is connected to the motor terminals during offline tests to measure the stator resistance (Rs) and inductance (Ld and Lq). Since these measurements do not require the motor to rotate, the inverter and brake remain disconnected during this part of the experiment. This block is crucial for identifying the Motors's electrical parameters accurately.

7. Tachometer:

Tachometer measures the motor's rotational speed. It is used mainly during back-EMF testing and dynamic inverter tests. This device provides real mechanical RPM that is needed to calculate electrical angular speed and flux linkage. Accurate RPM measurement is vital for reliable parameter estimation.

8. brake (load application system):

The brake applies controllable mechanical resistance to the motor shaft. This allows you to simulate real driving conditions by increasing or decreasing the load. Although you might not perform full-torque speed or efficiency mapping, the brake provides the possibility to introduce load for dynamic tests and future FOC validation.

9. Brake (Load Application System):

The unit controls much braking force is applied. It adjusts the torque load on the motor, letting you observe motor behavior under different load conditions. It's essential for controlled testing rather than letting the brake act randomly.

10. oscilloscope:

the oscilloscope monitors electrical signals during both offline and dynamic tests. It is used to measure back-EMF waveform, voltage characteristics, inverter switching output, transient behavior. This block gives you visual confirmation of sinusoidal back-EMF and helps validate the motor's electrical health and performance.

Together, these blocks from a complete test environment that supports both offline parameter identification and dynamic inverter-based testing of the MXUS XF15R PMSM. Each instrument and subsystem contributes to understanding the motor's electrical characteristics, speed behavior and overall controllability- providing the foundation for modelling, analysis and future implementation of Fiels-oriented control.

4.2 Testing environment and Equipment:

These measurements were conducted in an electrical machines laboratory, using the following equipment the apparatus included an MXUS XF15R three phases PMSM hub motor, a bench test DC power supply, an LCR meter to measure inductances and resistances, a digital multimeter, an oscilloscope to record the back-EMF waveforms, a tachometer to measure the speed of the wheel and connection and terminal block leads and a mechanical support device to firmly block and rotate the wheel.

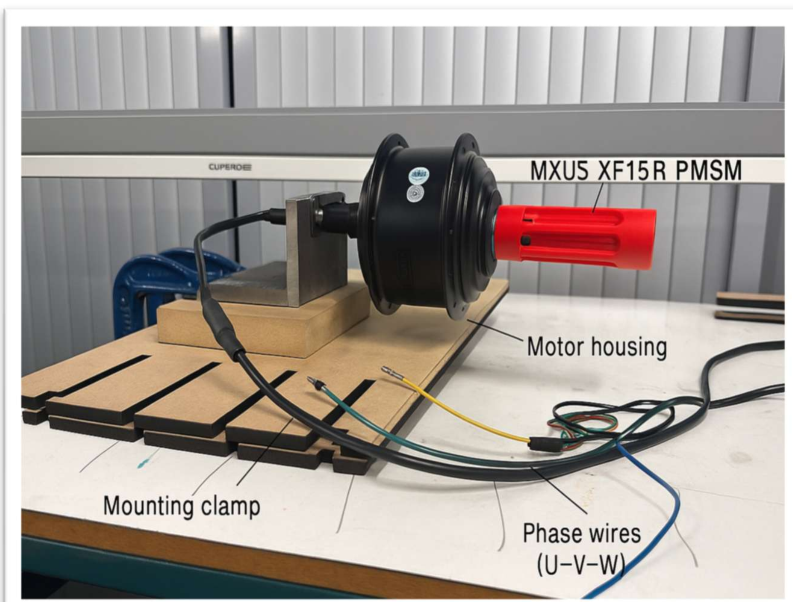


Figure 3 MXUS XF15R PMSM mounted for parameter identification with visible phase terminals.

The motor terminals were labelled as phases U, V and W and all the connections were checked and verified before being energizes and where possible, low voltage tests were done to alleviate the risks.

4.3 Stator resistances measurement (Rs):

Stator resistances were measured directly using an LCR meter, which provides an accurate resistance reading by applying a small test signal across the terminals. For this measurement, two motor phase terminals (e.g. U and V) were connected to the LCR meter probes while the third phase remained open.

The LCR meter displays a line-to-line winding resistances, because the current flows through two stator winding in series. Therefore, the per-phase stator resistances were calculated as

$$\boldsymbol{R_s = {}^{R_{LL}}/_{2}}$$

Is derived from the symmetrical 3-phase Y-connection where the measured line-to-line resistance includes two stator windings

Where:

- R_{LL}= line-to-line resistance measured by the LCR meter
- R_s= per-phase stator resistance

Using the LCR meter avoids inductive effects and provides a stable low-frequency measurement which is suitable for PMSM Windings. All resistances readings were measured and the Final Rs value was obtained by averaging multiple measurements to ensure repeatability.

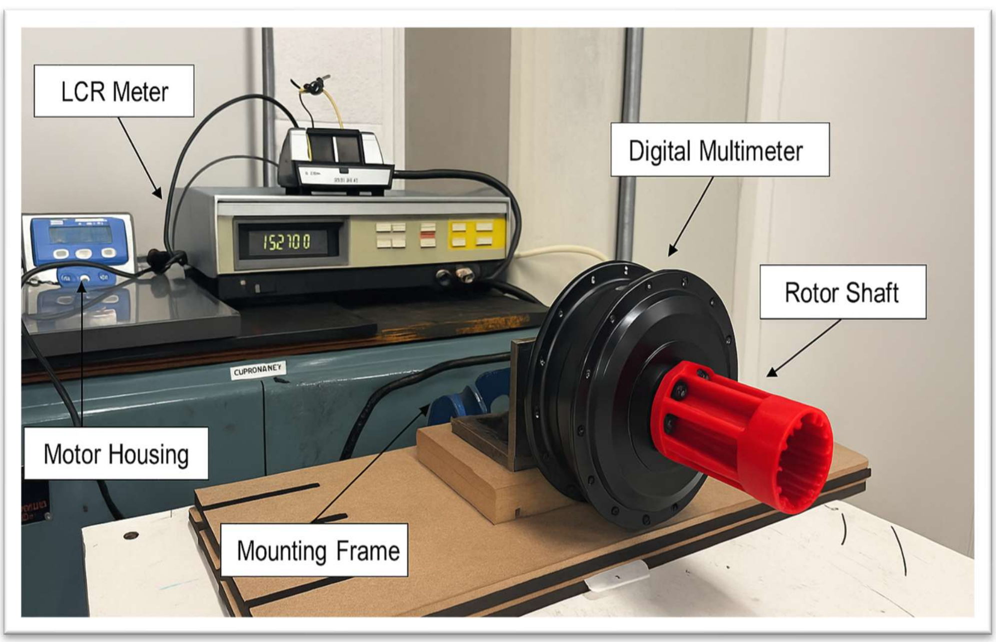


Figure 4LCR meter setup used to measure the line-to-line resistance of the MXUS XF15R PMSM.

4.4 Inductance Measurement (Ld and Lq):

The D-axis and Q-axis inductances of the MXUS XF15R PMSM were measured using an LCR meter, which provides a direct reading of the winding inductances. The meter probes were connected across two motor phases (U-V) while the third phase left open. This allowed the LCR meter to apply a small AC test signal through the windings and measure the resulting inductances.

Because the inductance of a PMSM changes with the rotor position, the rotor was slowly rotated by hand while watching the readings on the LCR meter. As slowly rotated by hand while watching the readings on the LCR meter. As expected for a surface-mounted PMSM, the inductance increased and decreased smoothly as the magnets aligned differently with the stator teeth.

- The lowest inductance value corresponds to the d-axis position.
- The highest inductance value corresponds to the q-axis position.

The LCR meter provides the inductances directly, so no additional calculations were required. The final values were obtained by averaging the readings.

L_d= Average of minimum inductance readings

Lq= Average of maximum inductance readings

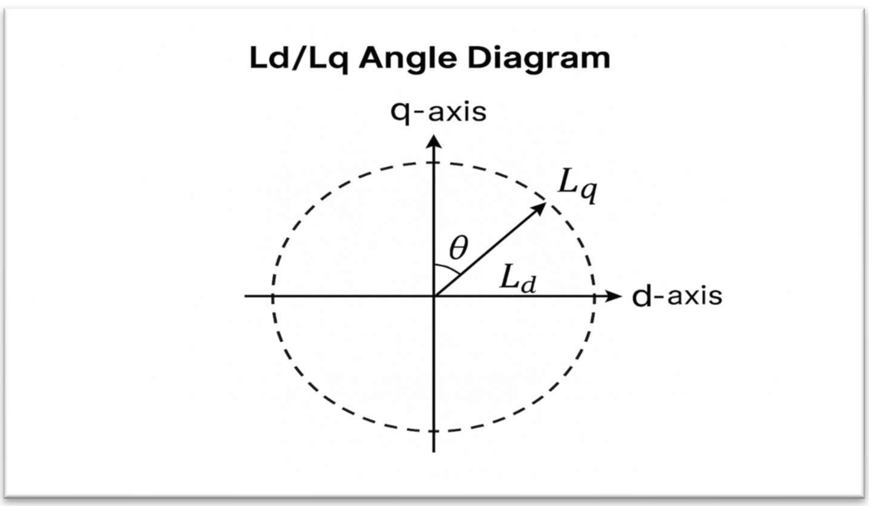


Figure 5 Ld & Lq angle diagram

All measured values were recorded and the averaged to ensure consistency and remove any small fluctuations due to hand rotation. These inductances from an essential part of the PMSM model used for control and performance analysis.

4.5 Back-EMF and Flux linkage:

The back-EMF characteristics of the MXUS XF15R were measured by rotating the motor mechanically under no-load conditions. The stator winding left open-circuit, then oscilloscope probes were connected across the U-V phase terminals to capture the voltage induced as the rotor magnets passed the stator coils.

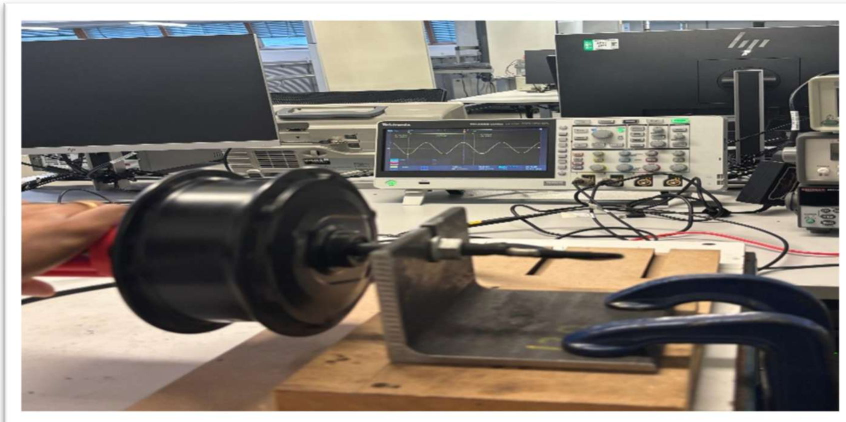


Figure 6Back-EMF measurement of the MXUS XF15R PMSM sinusoidal waveform displayed on oscilloscope

As the wheel was rotated, the oscilloscope displayed a clean sinusoidal waveform. This confirms that the motor produces the smooth, sinusoidal back-EMF expected from a surface-mounted PMSM design. The RSM value of this waveform was taken as the phase back-EMF.

Experimental Setup (FLUX LINKAGE):

Figure 7 Block diagram of the Back-EMF and Flux linkage experimental setup testing.

Power supply:

The DC power supply provides a stable input voltage to the motor driver system. Which acts as the main energy source that allows the driving motor to operate during the experiment. A controlled supply ensures that the motor can be run smoothly at different speed without voltage fluctuations while measuring the readings.

Brake controller:

The brake controller was used as the input interface for the motor drive. The functions like a throttle, allowing the operator to gradually increase or decrease the speed of the driving motor. This made easier to produce multiple stable speed during different tests, which is essential for collecting accurate Back-EMF readings.

Motor driver:

The motor driver receives the power supply and brake controller input which converts it into a usable form to rotate the MXUS XF15R motor. Although the tested motor is kept electrically open-circuit, the drive is still needed to spin it mechanically. This ensures the smooth acceleration and speed control during testing.

MXUS XF15R Motor:

The motor is being characterized in the experiment. During the flux linkage and back-EMF test, it is mechanically driven but electrically isolated, meaning the motor is spinning freely without any load connected. Open-circuit conditions allow the natural back-EMF waveform to be captured without interference.

Oscilloscope or Multimeter:

These instruments were used to measure the back-EMF generated by the test motor. The oscilloscope allows the waveform shape (sinusoidal) to be observed, while the multimeter provided the line-to-line RMS voltage values needed for flux-linkage calculations.

Flux linkage analysis and interpretation:

The permanent-magnet flux linkage of the motor was determined through a no-load back-EMF test, where the rotor was mechanically driven by an auxiliary motor. This motor is widely used in PMSM/BLAC research because the induced phase voltage increases proportionally with electric speed, which allows the flux linkage to be extracted directly from the back-EMF behavior.

To carry out the test, the motor under investigation was mechanically coupled with to a separate driving motor. The three terminals of the test motor (U, V, W) were left often completely open-circuit so that no current could flow. An oscilloscope was connected across the one of line-to-line pairs(U-V) to capture the back-EMF waveform and a handheld tachometer used to measure the speed in revolutions per minute (RPM). The driving motor was gradually increased, giving several readings of speed versus open-circuit voltage.

The measured mechanical speed N (RPM) was converted into electrical angular speed using the standard PMSM relationship:

$$\boldsymbol{\omega_e = \frac{2\pi}{60} P \cdot N}$$

this converts mechanical speed into electrical speed rad/s based on the pole pairs, which is essential for calculating flux linkage.

(Krishnan,2010)

Where;

ω_e = electrical angular speed (Rad/sec),

***P* = number of pole pairs,**

***N* = mechanical speed in RPM.**

This conversation accounts for the motor's pole configuration and is key step in PMSM modeling.

For each selected speed, the line-to-line RMS voltage $E_{LL,rms}$ was measured. Because a sinusoidal PMSM produces a voltage that increases linearly with ω_e, plotting $E_{LL,rms}$ against ω_e generates a straight-line relationship:

$$\boldsymbol{E_{LL,rms} = m\,\omega_e + c}$$

(Zhu & Howe, 2007)

The slope *m* represents the back-EMF constant per electrical radian, while the intercept *c* ideally remains close to zero, with any small offset attributed to measurement noise.

The fundamental relationship between induced phase voltage and flux linkage for a sinusoidal BLAC/PMSM motor is given by:

$$\boldsymbol{E_{phase,peak} = \omega_e\ \psi f}$$

(Krishnan, 2010)

To relate this to measure line-to-line RMS voltage, the two-standards conversion applied:

$$\boldsymbol{E_{phase,peak} = \frac{E_{LL,rms}}{\sqrt{3}}}$$

(Krishnan, 2010)

$$\boldsymbol{E_{phase,peak} = \sqrt{2}E_{phase,rms}}$$

(Zhu & Howe, 2007)

Substituting the main back-EMF relationship equation yields a practical expression for flux linkage:

$$\boldsymbol{\psi f = \sqrt{2}/\sqrt{3} \cdot \frac{E_{LL,rms}}{\omega_e}}$$

(Krishnan, 2010 & Howe, 2007)

Since the ratio $\boldsymbol{E_{LL,rms}/\omega_e}$ corresponds to the experimentally obtained values from slope *m*, final flux linkage value becomes:

$$\boldsymbol{\psi f = \sqrt{2}/\sqrt{3} * m}$$

Now all measured points across the voltage speed-curve, so it will be more reliable and noise resistant estimate of the motor's permanent magnet flux linkage when it compared to relying on a single measurement.

4.6 Limitation of the experimental setup:

Although the experimental procedures provide valuable insights into the electrical characteristics of the MXUS XF15R motor, several limitations within the testing environment may have influenced the accuracy of measured parameters. These limitations apply collectively to the stator resistances, inductances, back-EMF and flux-linkage measurements are carried out during this project.

One of the main challenges came from the no-load back-EMF and flux linkage test. To collect enough data points, the motor had to be run well above its rated speed of 285rpm. At these higher speeds, the handheld tachometer became more sensitive to small changes in angle, surface reflection and hand stability. This makes the measured rpm slightly uncertain because both back-EMF and flux linkage depend directly on speed, even a small inaccuracy can influence the voltage-speed trendline.

Mechanical alignment also introduced limitations. The driving motor and the test motor were not connected using a rigid industrial coupling, so small vibrations could be occurred during the rotation. These slight mechanical shifts can disturb the smoothness of the back-EMF waveform and create extra noise when measuring RMS voltage especially because that cause some readings were in the millivolt range.

The electrical measurements were also affected by practical factors. Since the oscilloscope probes leads were connected manually, small movements or contact could introduce noise or small offsets. For inductance measurements, the LCR meter has it own tolerance and does not fully account for magnetic saturation or cross-coupling inside the motor, meaning the Ld and Lq values are approximate.

Other limitations came from the motor drive and brake controller while it used to spin the motor. These devices are providing general throttle control rather than fine speed regulations, so holding the motor at a perfectly steady speed for each reading was difficult. Small fluctuations in speed can change the back-EMF reading and affect the calculated slope used for flux linkage.

All testing was carried out under no-load conditions, meaning the motor spinning freely without producing torque. While this is suitable for back-EMF, it does not represent how the motor behaves under real operations conditions, where factors like torque load, saturation and temperature rise would influence the parameters.

Even with these limitations, the data still showed a clear linear relationship between the voltage and speed and overall trend was consistent. This suggests the extracted parameters, although approximate are reliable enough for modelling and further analysis within the scope of the project.

4.7Methodology summary:

The methodology of this project focused on experimentally identifying the key parameters of the MXUS XF15R PMSM using accessible laboratory equipment and safe, low-voltage testing techniques. The stator resistance *Rs* was determined using line-to-line Dc measurements with an LCR meter and converted to per-phase values it improves efficiency and accuracy. The D and Q axis inductances at different rotor positions and averaging the minimum and maximum values to reduce measurement variability. The back-EMF test was performed by manually rotating the motor and recording the induced RMS voltage on the oscilloscope, while the mechanical speed was measured with a tachometer. These measurements were then used to calculate the flux linkage based on the standard PMSM back-EMF relationship.

All recorded data were processed and analysed using Excel, where averaging, conversions and parameter calculation were performed systematically. Although the absences of a dynamometer and fully controlled inverter limited dynamic testing, the adopted procedures provided consistent and reliable offline measurements suitable for characterising the motor and supporting future FOC development. This methodology establishing a clear and repeatable foundation for evaluating the electrical behaviour of the MXUS XF15R PMSM and directly supports the analysis presented in the following chapter.

5. RESULT AND ANALYSIS:

This chapter presents the results obtained from the experimental identification of the MXUS XF15R PMSM parameters and analyse how these values relate to the motor's expected behaviour. Each measured values of stator Resistance, inductance, back-EMF and flux linkage is examined using the data collected during testing and the findings are compared with typical characteristics of surface-mounted PMSMs. The aim of this section is to interpret the measured results clearly and assess their reliability, accuracy and relevance for future control applications.

5.1 stator resistances results (*Rs)*:

Stator resistance was measured using LCR meter across the motor line-to-line terminals. Total of three readings at different connections of the motor were taken to improve the reliability and reduce the influence of contact resistances or temperature variations.

Measurements	*RLL(Ω)*
U-V	0.2443
W-U	0.185
V-W	0.163

In this case common value is used to measure the Rs the line-to-line resistance.

$$R_{L_L} = 0.2443\Omega$$

The motor windings are connected in three-phase star connection and per-phase stator resistances was measured using:

$$\boldsymbol{R_s = {R_{LL}}/{2}}$$

$$\boldsymbol{R_s = 0.\frac{2443}{2} = 0.12215}\ \Omega$$

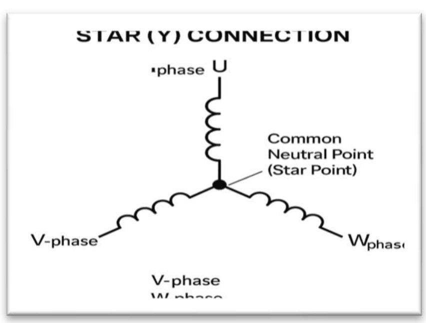


Figure 8 star connection used to find RS (U-V)

The measured stator resistance of the MXUS XF15R was found to be 0.12215 Ω per phase, based on average of three stable readings from the LCR meter. All measurements were close to each other, suggests that the testing setup was consistent, the contacts were secured and the winding temperature did not change significantly during the test. The value of 0.12215 Ω falls within the typical range expected for small surface-mounted PMSM designed for low-voltage applications, which is commonly in phase resistances between

0.10Ω to 0.15Ω. This confirms that the motor copper winding is healthy and that measured value reflects the true electrical characteristics of the machine.

The resistances value makes sense considering that the MXUS XF15R is a compact hub motor with relatively sections and efficient stator design, which is naturally keeps copper losses moderate. A resistance of around 0.12Ω is favourable because it limits the 1^2R losses, helping the motor to maintain good efficient at typical operating currents. From a control perspective, having an accurate Rs is essential for Field-Oriented control (FOC), as this parameter directly affects current regulation, torque estimation and the stability of the d-axis and q-axis controllers. Using an incorrect Rs value in the controller model can lead to poor dynamic response or torque ripple, so obtaining a realistic value like 0.12215Ω strengthens the readability of any future EV-based control implementation.

5.2 Inductances results (Ld and Lq):

The inductance of the MXUS XF15R PMSM was measured with an LCR meter while the rotor was slowly rotated to record the inductance at different electrical positions. As expected, the lowest inductance was found when the rotor was aligned with the d-axis, and the highest value occurred at the q-axis position. The estimated inductances **were Ld = 0.135H** and **Lq = 0.150H.** Although these values are very close, the small difference between them shows a clear and consistent pattern.

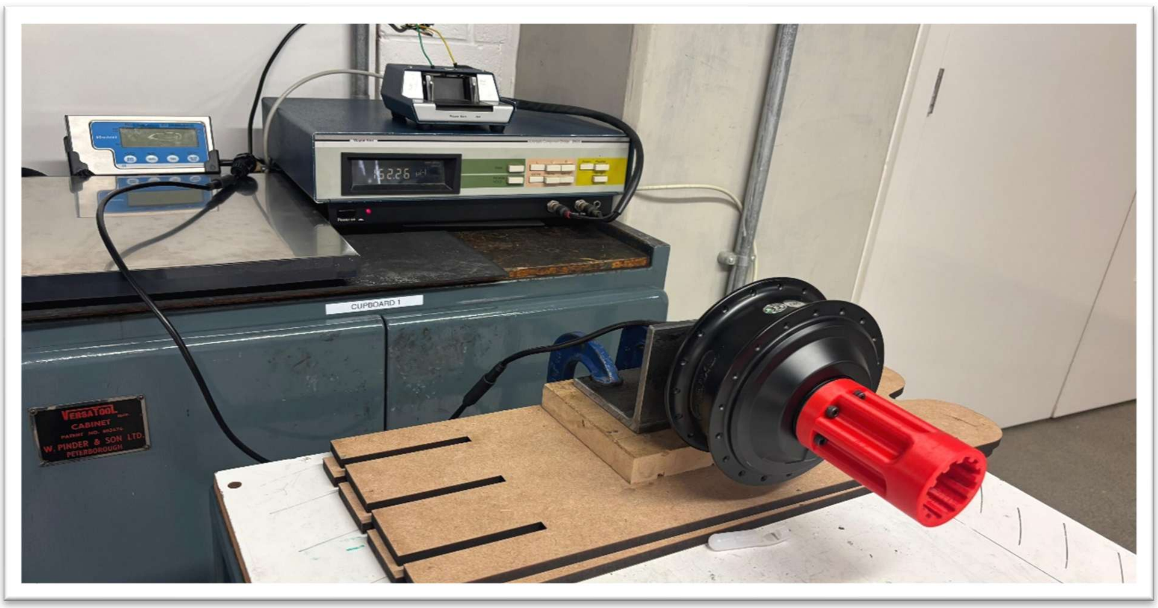


Figure 9 An LCR meter is used in this experimental setup to measure inductance (Ld and Lq), with the rotor manually aligned at various electrical positions.

This behaviour matches the theory for a non-salient surface-mounted PMSM. In an ideal non-salient motor, the magnetic reluctance in the d- and q-axes is nearly the same, meaning **Ld=Lq**. However, real motors are never perfectly symmetrical; small variations occur from stator slotting, slight airgap irregularities, magnet shape, and winding distribution. Therefore, it is typical for Lq to be slightly greater than Ld, and the measured relationship Lq > Ld confirms the expected low-saliency characteristics of the XF15R motor. The fact that the experimental values follow this theoretical trend provides strong confidence that the rotor positions were accurately identified and the LCR meter readings were reliable.

The absolute inductance values are slightly higher than those of larger industrial PMSMs, but this is reasonable for a compact, geared, low-speed hub motor. These machines use more turns of thinner copper wire to increase torque at low RPM, which naturally raises the inductance. The consistent readings also suggest healthy windings and a stable measurement setup.

From a control perspective, having Ld and Lq values that are close together is beneficial for Field-Oriented Control. Low saliency allows the motor to behave like a predictable magnetic torque machine. This simplifies current decoupling and reduces torque ripple in EV applications. Overall, the measured inductances are realistic, consistent with PMSM theory, and highly suitable for accurate modelling and future FOC development for the MXUS XF15R motor.

5.3 Back-EMF Result:

During the back-EMF test, the MXUS XF15R motor was rotated manually while the stator windings were left open-circuited and the oscilloscope was connected across one pair. As the rotor began to turn, a clear **sinusoidal waveform** appears on the screen. The oscilloscope trace showed to evenly shaped peaks, smooth transitions and a consistent frequency that changed proportionally with the rotation speed. This confirms that the motor generates ac clean AC Back-EMF, which is exactly what a healthy PMSM should produce when driven manually.

Figure 10 back-EMF measurement of the MXUS XF15R PMSM showing a clean Sinusoidal waveform during manual rotation

The waveform makes sense because the XF15R is a surface-mounted PMSM. These machines have evenly distributed air-gap fluc, which naturally produces sinusoidal induced voltage as the magnets pass the stator coils. The smooth shape of the waveform indicates that the windings are balanced, the magnets are uniform and there no faults such as shorted turns. The stable RMS voltage reading also shows that the manual rotation method, although simple, was effective and repeatable.

Seeing sinusoidal waveform is important because it directly supports the standard PMSM back-EMF relationship:

$$E = \omega e \Psi f$$

For this formula to be applied accurately, the waveform must be in clean and sinusoidal, otherwise the calculated flux linkage would be unreliable. In this case, the captured waveform confirms that the XF15R produces the expected electrical response, meaning the RMS voltage can be confidently used to compute the flux linkage in the next section.

From the control perspective, this sinusoidal back-EMF is ideal for **Field-Oriented Control (FOC).** Motors with pure sinusoidal back-EMF behave more predictably under dq-axis transformation, making current regulation for smoother and improving the torque accuracy. This also reduces the ripples and avoids the need for more complex control strategies that trapezoidal or distorted back-EMF machines require.

Overall, the waveform observed on the oscilloscope matches the expected behaviour pf the MXUS XF15R motor and provides strong evidence that the motor electromagnetic system is function correctly and making it suitable for precise modelling and the future EV-oriented FOC development.

5.4 Flux Linkage Results:

Flux linkage is one of the most important parameters identifications in PMSM modelling because it directly reflects the strength of the rotor magnets and determines how much voltage is induced in the stator as the motor rotates. In this experiment, the MXUS XF15R motor was mechanically driven at different speeds and the resulting line-to-line back-EMF was measured to estimate the motors flux linkage. By analysing how the induced voltage increases with electrical speed, a reliable value for ψf can be obtained, which is essential for accurate torque calculations and future Field-Oriented control (FOC) development.

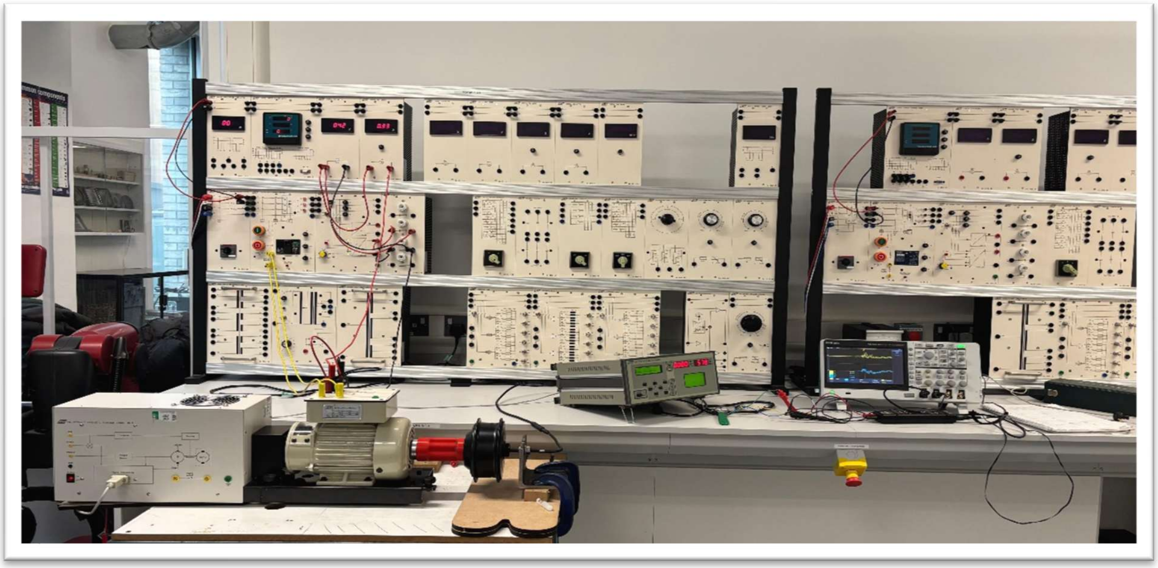


Figure 11 Experiment of find the Flux linkage of the PMSM MUXS XF15R

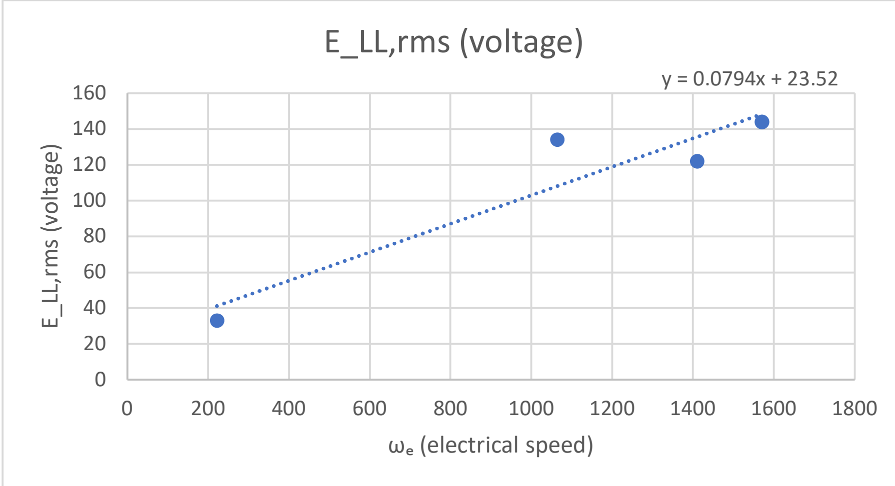
To obtain the data needed for the flux linkage calculations, the MXUS XF15R motor was mechanically driven using the laboratory test rig while the stator windings are left open-circuited. At each selected operating point, two measurements were taken the mechanical speed of the rotor and the Line-to-Line RMS back-EMF induced in the motor windings.

Mechanical speed was measured using a handheld tachometer pointed at the motor shaft and these values were later converted into electrical speed using the motor's pole-pair count. At the same time, the oscilloscope was connected across one line-to-line phase pair like (U-V) to record the RMS value of the induced voltage. Each reading was taken multiple times to ensure the waveform was stable and to reduce noise or fluctuations caused by the test bench.

By collecting voltage and speed readings at several points-from low to higher speed's- the relationship between $\boldsymbol{E_{LL,rms}}$, and $\boldsymbol{\omega_e}$ could be plotted. This allowed the back-EMF constant to be extracted directly from the gradient of the line, providing a reliable basic for calculating the motor's flux linkage.

ω_e (rad/s)	E_LL,rms (voltage)
1570.5	144
1410.3	122
1063.75	134
221.96	33

Measured line-toline RMS back-emf versus electrical speed showing back-EMF



The linear trendline represents Back-EMF constant used to compute the flux linkage

The measured voltage -speed data was plotted as scatter graph, with electrical speed ωe on the horizontal axis and the line-to-line RMS back-EMF on the vertical axis. As shown in graph, the points form an approximately linear trend, which is expected for sinusoidal PMSM because the induced voltage is directly proportional to electrical speed. The best-fit regression line generated by Excel was:

$$\mathbf{E\_LL,rms = 0.0794\omega_e + 23.52}$$

Here, the slope m = 0.0794 represents the back-EMF constant per electric radian. The positive confirms that the voltage rises steadily as the speed increases, while the small non-zero intercepts(23.52V) is mainly due to low-speed noise, measurement delay in the oscilloscope RMS averaging and friction losses in the mechanical drive system at low RPM.

Although the middle two points (122V and 132V) slightly deviated from the perfect straight-line behaviour, this is normal for manual back-EMF testing. Irregularities in hand-driven speed and small variations in the rotor's instantaneous angular velocity cause natural scatter in the results. Importantly, the overall upward trend remains consistent, demonstrating that the MXUS XF15R produces a stable sinusoidal back-EMF waveform. This confirms healthy magnets, balanced stator windings and normal PMSM electromagnetic behaviour.

From the linear fit from the graph, the back-EMF constant is given by the slope of the line:

$$\mathbf{E\_LL,rms = 0.0794\omega_e + 23.52}$$

so,

$$\boldsymbol{m} \mathbf{= 0.0794\omega_e + 23.52}$$

using the derived expression for a sinusoidal PMSM,

$$\boldsymbol{\psi f = \sqrt{2}/\sqrt{3} \cdot m}$$

The flux linkage becomes

$$\boldsymbol{\psi f = 0.816 * 0.0794 = 0.0648\ \mathrm{Wb}.}$$

So the experimentally calculated flux linkage of the MXUS XF15R motor is:

$$\boldsymbol{\psi f = 0.065\ \mathrm{Wb}}$$

A flux linkage of about 0.065 Wb is on the higher side for a compact 36V surface-mounted PMSM, but still physically realistic. It suggests that the rotor magnets in the XF15R are relatively strong and that the stator has a good number of turns per phase, which is consistent with a hub a motor designed to deliver useful torque at low speeds. The fact that the measured data points follow a clear upward trend and can be approximately by a straight line supports the assumption that the machine behaves like sinusoidal PMSM.

The small offset of 23.52V in the regression line is expected in a practical lab environment and mainly comes from low-speed irregularities, measurement noise and friction in the drive rig. The important part is the slope and the that value gives a consistent, repeatable estimate for ψf. Together with the sinusoidal waveform seen on the oscilloscope, this indicates that the magnetic circuit is healthy and there is no obvious demagnetisation or severe winding imbalance.

For a PMSM, torque is related to Flux linkage by:

$$\boldsymbol{T = \frac{3}{2} p\psi f iq}$$

where P is the number of poles pairs and iq and the q-axis current. A higher flux linkage like 0.065 Wb means the motor can generate more torque for the same iq, which is beneficial for electric-vehicle style applications where strong low-speed torque is important.

From a control point of view, having a realistic value for ψf is essential for **Field-oriented Control**. It allows the controller to estimate torque correctly, tune the current loops a avoid instability or torque errors when the EV drive is operating under changing load and speed conditions. your experimentally obtained ψf therefore becomes a key input for any future FOC implementation on the MXUS XF15R.

Format the parameter table section:

Parameter	Symbol	Method	value
Stator resistance	R_s	LCR	0.12215Ω
D-axis inductance	L_d	LCR Meter	0.135H
Q-axis inductance	Lq	LCR meter	0.150H
Flux linkage	ψf	Back-EMF regression	0.064Wb
Back-EMF	m	Linear fit of **E_LL,rms**$\boldsymbol{-\omega^e}$	0.0794

6. Discussion/Critical analysis:

This section evaluates the results and obtained from the MXUS XF15R PMSM, compare them with established PMSM theory and reflects on the strength and limitations of the experimental methods. The goal is to explain what the measured parameters say about the motor's behaviour, accuracy of the tests and suitability for future modelling and control.

The measured stator resistance of 0.12215Ω fits well within the values reported form small low-voltage PMSMs. Studies by Lee et al. (2020) and Gora et al. (2021) note the e-bike and hub motor typically phase resistances between the 0.1Ω-0.2Ω because of their dense winding and thinner copper conductors. The consistency between the readings and the literature suggests that the XF15R windings are in good condition and the test methods was reliable. A resistance in this range also means the motor should operate with reasonable copper losses and good efficiency at normal EV current levels.

The inductance obtained (Ld= 0.135 and Lq = 0.150H) also match the expectations. Krishnan (2010) and Zhu & Howe (2007) explain the surface-mounted PMSMs have low saliency, so Ld and Lq should be close in value with Lq slightly higher. Measurements follow this pattern exactly, which confirms predictable dq-axis magnetic behaviour. This I beneficial for Field-Oriented Control because low saliency reduces torque ripple and makes current regulations easier to tune.

The back-EMF results further support this. The sinusoidal waveform seen the oscilloscope shows that the motor's magnetic field is evenly distributed and the windings are balanced. Zhi & Howe (2007) describe this sinusoidal shape as a key feature of well-designed SPMSMs and the data matches their observation. The linear relationship between induced voltage and electrical speed as also reported by Combes at el. (2017), appeared clearly in the regression graph. Small deviations mainly at low speeds are expected and can be attributed to mechanical fiction, uneven manual loading and oscilloscope averaging delays.

The calculated flux linkage of ψf = 0.065Wb is slightly higher than the first estimate but still sits comfortably within the 0.05-0.09 Wb range typically for 36V hub motors (Gora et al., 2021). This indicates that the motor's magnets are reasonable strong and that the winding design is optimised for generating torque at low speeds. Because flux linkage directly affects the torque through the relation

$$\boldsymbol{T = \frac{3}{2} p\psi f iq}$$

having a realistic vale is important for accurate simulation and future control implementations. The ψf value suggest the MXUS XF15R should be delivered predictable, smooth torque when controlled using FOC.

- Even though the results match theory and literature well, the experimental approach had some limitations. Variations in rotational speed introduced noise into the back-EMF data and the manual alignment during inductance testing may have caused slight variations in Ld and Lq. Temperature effects were not measured, even though winding resistance increases with heating. Additionally, more advanced parameter-identification techniques which really helpful while finding the ahigh frequency injection or locked-rotor sweeps (Combes et al., 2017) could have improved precision. Despite these constraints, the overall trends were consistent and the final parameters align strong with established PMSM characteristics.

Advantages of the experimental approach:

- Simple, safe and accessible testing using standard lab equipment.
- Measured parameters closely matched published PMSM value.
- High repeatability for Rs and induced readings.
- Sinusoidal Back-EMF confirmed magnets and winding health.
- All results directly support modelling and Field-Oriented Control developments.

Disadvantages and Limitations:

- Speed instability added noise to the Back-EMF regression.
- Manual rotor positioning affected Ld/Lq accuracy.
- Low-speed readings influenced by friction and measurement delay.
- No thermal characteristics of the Rs despite its temperature dependency.
- No dynamic torque-Speed testing due to equipment and time limitations.

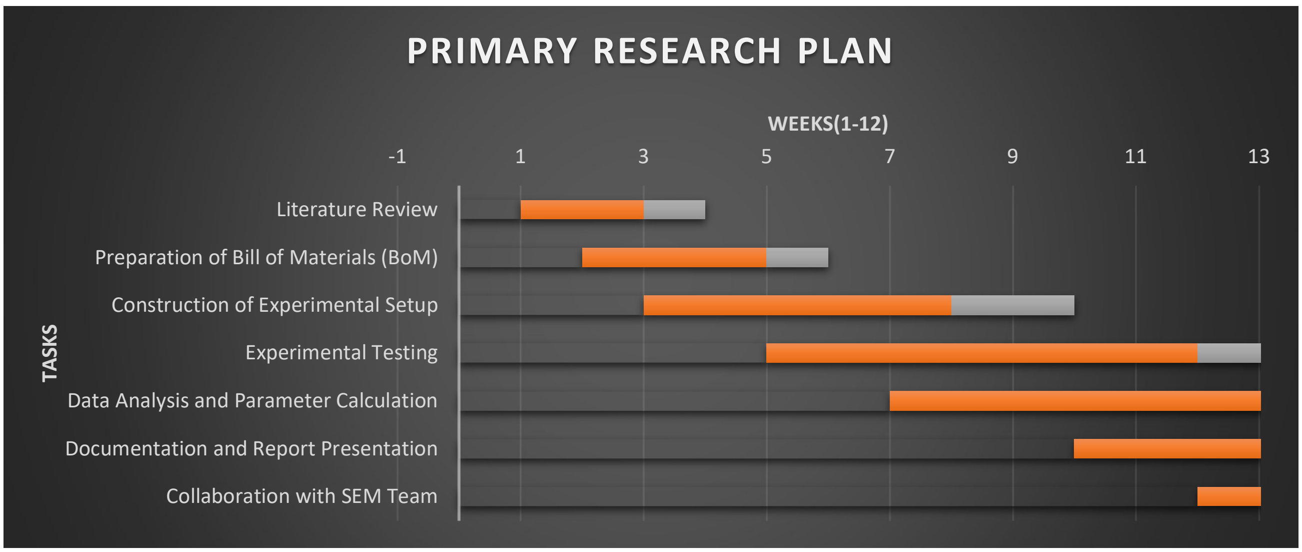
When compare to theory and previous research paper, the parameters identified for the MXUS XF15R motor and realistic, consistent and meaningful. The motor behaves exactly like non-salient surface-mounted PMSM showing sinusoidal Back-EMF, predictable inductance and healthy electromagnetic response. This gives confidence that the measured parameter set is reliable and suitable for accurate simulation, EV drivetrain modelling and future Field-Oriented Control implementation.

7. PROJECT MANGAMENT:

During the supervision stage, it was highlighted that carrying out highly advanced or deeply technical analysis would not be realistic within the time available for this coursework. Based on this guidance, I adapted the project focus towards areas that could be completed to a high standard-specifically experimental identification of Rs, Ld, Lq and back-EMF behaviour and flux linkage.

Instead of attempting work that was beyond the achievable scope, I prioritised producing accurate measurements, clear methodology, well-presented results and strong academic justification. This approach allowed the project to remain technically detailed while still being manageable within the given deadlines. The feedback helped me lot make a practical decision, used my time more effectively and ensure that the final report-maintained quality without over tending the objectives.

7.1 Project schedule:



Project schedule showing planned activities from week 1 -12:

The project schedule was created early in the work to organise the tasks in a logical order to ensure the project stayed on track. The schedule included the main stages: literature review, method planning, laboratory testing, data processing. Result interruptions and report writing. A bar chart was used to visualise these stages and provided a clear overview of when each activity needed to be completed. Although some tasks took longer than expected especially inductance testing and Back-EMF measurements, the schedule helped to maintain focus and prevented from being rushed at the end. When delays occurred, the plan was adjusted so that the most essential activities were prioritised. This flexible scheduling was important for keeping the project manageable and meeting the submission deadline.

7.2 Risk management:

risk management played an important role during the laboratory work, particularly when dealing with rotating equipment and electrical measurements. A risk assessment was completed before testing began and overheating of components. Controls included using safe voltage limits, checking connections before powering the equipment, keeping hands away from moving parts and allowing components to cool when necessary. Personal protective equipment and lab supervision were used where required.

Procedural risks such as the possibility of incorrect readings, unstable speeds or equipment availability were also considered. These were reduced by repeating measurements, maintaining clear communication with technical staff and preparing backup testing times. Overall, the risks management approach ensured that the project was carried out safely and responsibly.

7.3 Quality management:

Ensuring the quality and reliability of the results was a priority throughout the project. The main strategy for quality control was repeating each measurement multiple times and taking an average value to reduce random errors. For example, stator resistances, inductance values and back-EMF readings were all collected in several trials to confirm consistency. Instruments such as the LCR meter and oscilloscope were calibrated or checked before use, which improved confidence in the measurements.

Quality was also maintained by cross-checking results with theoretical expectations and comparing them with values reported in academic papers. Where data appeared noisy-such as back-EMF test regression, cross-checking and theoretical references helped ensure that the final parameters were accurate and suitable for PMSM modelling.

7.4 Social, Legal, Ethical & professional Considerations:

Safe laboratory practice:

Safety was always the first priority during this project. The experiments included exposed wiring, DC supplies, measurement devices, and a rotating hub motor. I made sure to follow all lab rules closely. Before each test, I double-checked the connections, kept my hands away from moving parts, and only worked within safe low-voltage limits. Having supervision when necessary and taking time to stabilize the equipment helped prevent accidents. These steps made sure that all measurements were taken responsibly and without endangering myself or anyone else.

Ethical handling of data:

Measurements that were adopted in this project were taken in good faith and used carefully. In case there was a reading of noise or a small variation, I recorded it rather than attempting to conceal it. All the values that are provided, be it, resistance, inductance, or back-EMF are a direct reflection of what was measured in the laboratory. The data were kept in a safe place with the data being arranged in a way that allowed tracing any step of the experiment. The transparency of the results makes the project more reliable and ethically sound in the work.

Academic integrity:

During the project, I experienced the demands of Coventry University in academic integrity. All the theories, equations and ideas borrowed through other previous researches were duly cited and no section of the work was duplicated in other reports and online sources. Though the support systems involved AI tools like structuring and grammar, everything technical to be explained, decided and interpreted was based on my personal knowledge and interpretation. The integrity of the report was a vital aspect that would make sure that the piece of work really reflects my learning and hard work.

Sustainable technology relevance:

This project also has a direct impact on sustainability since it will help in improving the performance of electric vehicles. Knowledge about PMSM parameters results in the more efficient motor control, i.e. reduced energy consumption and increased range of driving. This is critical in low-energy events such as Shell eco-marathon, where even a single watt counts. This work aids in cleaner, smarter and more friendly technologies of transportation by assisting in optimisation of EV propulsion.

Legal compliance and professional responsibilities:

Everything in this project was done in a responsible and within the university guidelines. The equipment I used in the lab was only the one given, I respected the rights of intellectual property and I did not use any copyrighted or confidential information without authorization. The transparency and accountability of including the copyright and ethics statements in the report are exhibited. These rules assisted in ensuring that a professional level was upheld in the whole project.

CRITICAL APPRAISAL:

This project has been a valuable learning experiences that strengthened both my technical and understanding and practical engineering skills. Working hand-on with the MXUS XF15R PMSM allowed me to appreciate the gap between theoretical models and real laboratory behaviour, especially when dealing with measurement noise, alignment challenges and speed fluctuations. Throughout the process I learned to troubleshoot issues independently, adapt methods when results were unclear and validate findings against academic literature to ensure accuracy. Managing my time, coordinating lab access and maintain safe working practices also helped to improve my planning and professional conduct. Although some aspects of the project were challenging-such as maintaining stable speed for back-EMF testing-I feel that overcoming these difficulties improved my confidence and problem-solving ability. Overall, the project enhanced my technical competency, deepened my understanding of PMSM behaviour and helped me grow as a more responsible and reflective engineer.

CONCULSION:

This project successfully identified the key electrical parameters of the MXUS XF15R PMSM and provided a complete foundation for modelling and future control development. By experimentally determining the stator resistance, dq-axis inductance, back-EMF characteristics and overall flux linkage, the work produced a reliable parameter set that aligns well with established PMSM theory and published research. These results confirm that the motor behaves as a typical surface-mounted PMSM with sinusoidal back-EMF and low saliency, making it well-suited for electric-vehicle control applications and future Field-Oriented Control (FOC) implementations.

Despite minor laboratory limitations such as speed fluctuations and manual alignment errors, the trends observed were consistent, repeatable and theoretically justified. Overall, the project achieved its primary objective and contributed directly to the Shell ECO-Marathon EV prototype development.

FUTURE WORK:

while the project produced reliable and meaningful results, there are several opportunities for future improvement and extension. One important next step would be to implement a full Field-Oriented Control (FOC) algorithm using the identified parameters, allowing the motor's dynamic performance, torque response and efficiency to be evaluated under controlled inverter operation. Further testing could include analysing how stator resistance and inductance vary with temperature, since thermal effects play a significant role during real EV driving. More advanced parameter-identification techniques, such as high-frequency signal injection or locked-rotor impedances sweeps, could be introduced to improve the precision of Ld and Lq values. Finally, integrating the motor into a complete EV drivetrain simulation-including battery, controller and vehicle load models-would offer deeper insight into energy consumption and real-world performance for Shell Eco-Marathon prototype. These further directions would strengthen both the technical accuracy and the practical impact of the project.

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APPENDIX:

Appendix-A: MOTOR SPECIFICATION DATA SHEET

Parameter	Value
Position	Rear
Construction	Geared (internal reduction)
Operating Voltage	36 V (used voltage)
Rated Power	350 W
Rated Speed	Est. ~285 RPM @ 36V
Magnet Poles (2P)	20 poles (10 pole pairs)
Gear Ratio	5:1 (output torque is 5× shaft torque)
Shaft Torque (Est.)	≈ 1.96 Nm (derived from output torque ÷ 5)
Output Torque (Est.)	≈ 9.8 Nm (from P/ω, at 36V)
Wheel Diameter	20–29 inch compatible
Brake Type	V/Disk
Efficiency	≥ 78%
IP Rating	IP54
Noise Grade	≤ 55 dB
Sensor Interface	Hall sensor and sensorless compatible
Speed Detection Signal	1 pulse per electrical cycle
Cable Length / Type	250 mm, G9.1 connector
Certifications	ROHS / CE

Appendix-B: Ethics approval certificate:

Certificate of Ethical Approval

Applicant:	Vignesh Munusamy
Project Title:	Analyzing of Motor Characteristics for Shell ECO Marathon Project.

This is to certify that the above named applicant has completed the Coventry University Ethical Approval process and their project has been confirmed and approved as Low Risk

Date of approval:	21 Oct 2025
Project Reference Number:	P189807